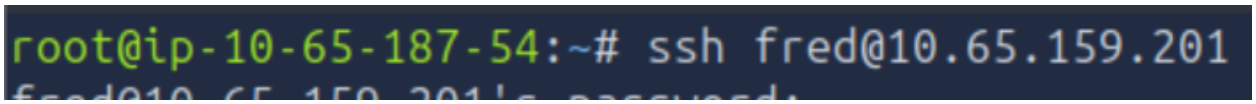
Alex Vong
Date: 10/02/2026

TryHackMe Room: Linux Server Forensics AV Write-up

Links that Help with Linux Investigations:

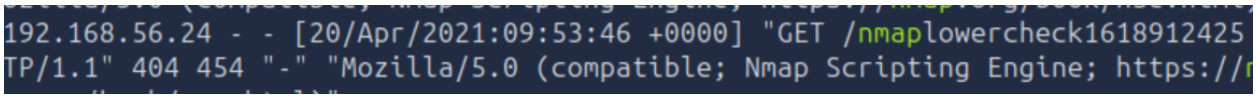
- https://www.dotlinux.net/linux-security-essentials/how-to-manage-and-identify-security-logs-on-linux/
- https://www.intramweb.com/linux-security/linux-server-security-commands/

Apache Log Analysis:



- SSH to this:
- Syntax for SSH: ssh username@IP_Address

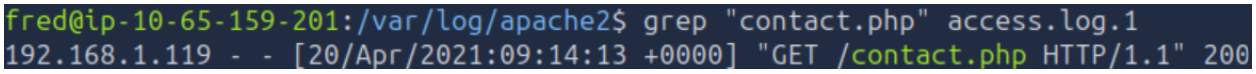
Apache is a popular open source web server.



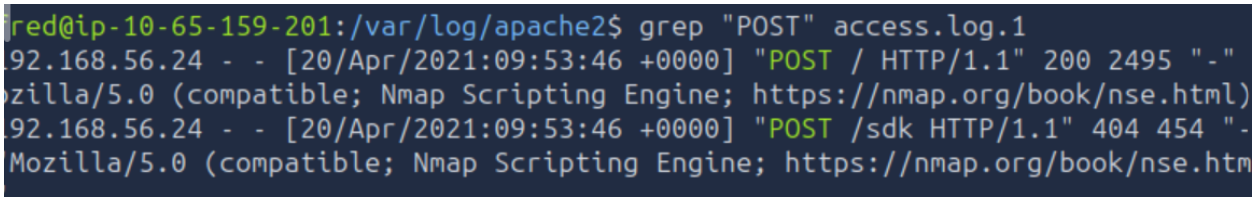
- Needed to find the NMAP requested file path
- Found this by doing **cat access.log.1 | grep "nmap"**

—----------

Web Server Analysis:



- This has a 200 HTTP code, which means it is working.



- This shows the IP address 192[.]168.56.24 is uploading files to server with index.php?

```
192.168.56.24 - - [20/Apr/2021:09:55:38 +0000] "HEAD /resources/2019/static/css/
adm.html HTTP/1.1" 404 140 "-" "DirBuster-1.0-RC1 (http://www.owasp.org/index.ph
p/Category:OWASP_DirBuster_Project)"
```

- Command **grep -i dirbuster acces.log.1** shows .24 address uploading.

```
fred@ip-10-65-159-201:/var/log/apache2$ cat access.log.1 | grep "error"
```

- **TRY THIS COMMAND AGAIN!**

```
fred@ip-10-67-152-176:/var/log/apache2$ cat error.log.1 | grep 'error'
[Tue Apr 20 09:53:46.649037 2021] [core:error] [pid 972] [client 192.168.56.24:5
5550] AH00135: Invalid method in request IDRT / HTTP/1.1
[Tue Apr 20 09:55:25.358036 2021] [autoindex:error] [pid 976] [client 192.168.56
.24:52085] AH01276: Cannot serve directory /usr/share/apache2/icons/: No matchin
g DirectoryIndex (index.html,index.cgi,index.pl,index.php,index.xhtml,index.htm)
 found, and server-generated directory index forbidden by Options directive
[Tue Apr 20 09:55:25.360043 2021] [autoindex:error] [pid 976] [client 192.168.56
.24:52085] AH01276: Cannot serve directory /usr/share/apache2/icons/: No matchin
g DirectoryIndex (index.html,index.cgi,index.pl,index.php,index.xhtml,index.htm)
 found, and server-generated directory index forbidden by Options directive
[Tue Apr 20 09:55:27.118986 2021] [php7:error] [pid 972] [client 192.168.56.24:4
4427] script '/var/www/html/products/contact.php' not found or unable to stat
fred@ip-10-67-152-176:/var/log/apache2$ whoami
fred
```

- Fred's name is here because this is his account. I did the command **whoami** to get that result. It took me a while to find out who did this.
- The account it's under made the exposure of the security notice on the server.

https://httpd.apache.org/docs/2.4/logs.html

- Apache log files support for errors.

—---------

Persistence Mechanisms I:

Remote File Inclusion (RFI) is a cyber attack where an attacker exploits a vulnerability in a web application to include malicious files from a remote server.

- By injecting URLs into input fields, attackers can execute arbitrary code on the target server, leading to potential system compromise or unauthorized access.

Can maintain Linux Persistence by:

1. cron
2. Services/systemd
3. bashrc
4. Kernel modules
5. SSH keys

The attacker used the command **sh -i** to establish a backdoor.

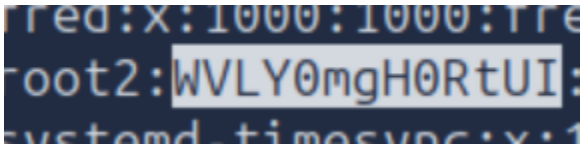
- This command invokes an interactive shell, often used for simple backdoor setups.
- Comes from the context of analyzing persistence mechanisms used by attackers on a compromised Linux server.
- This might be found embedded within scripts or cron entries used by the attacker to ensure they can regain access to the system even after a reboot or other cleanup efforts.

—----------

User Accounts:

Couple Linux Account Information:

1. **/etc/passwd** - contains the names of most of the accounts on the system. Should have open read permissions and should not contain password hashes.
2. **/etc/shadow** - contains names but should also contain password hashes. Should have strict permissions.



- Hash from doing command **cat /etc/passwd**

sudo awk -F: '/^root2/{print $2}' /etc/shadow — This prints the root2 account's password hash so you can inspect its prefix (e.g., 1, 6) to identify the hash algorithm before performing any authorized offline cracking or analysis.

- THM Echo AI agent says this…

```
fred@ip-10-67-152-176:~$ sudo awk -F: '/^root2/{print $2}' /etc/passwd
WVLY0mgH0RtUI
```

- I got the hash, but am not sure how to proceed from here yet.

https://github.com/sinfulz/JustTryHarder/blob/master/Priv_Esc

- This helps me by using the hash to make a password?

I followed this GitHub and created the password ‘mrcake’ using the hash.

```
fred@ip-10-66-180-146:~$ sudo echo "root2:WVLY0mgH0RtUI:0:0:root:/root:/bin/bash
" >> sudo /etc/passwd
fred@ip-10-66-180-146:~$ su root2
Password:
root@ip-10-66-180-146:/home/fred#
```

- I put ‘mrcake’ as the password when I logged into root2 user.

—----------

Deploy Second VM

I SSH into a new machine to deploy the second VM

Directory Brute-Forcing is requesting many folder and file names from a wordlist against a web server to find paths that aren't linked anywhere on the site's visible pages.

```
fred@ip-10-67-132-165:/var/log/apache2$ cat access.log.1 | cut -d ' ' -f6 | sort
 -u

"GET
"GXWR
"OPTIONS
"POST
"\x16\x03"
```

- Answer: GXWR is the non-standard HTTP request
    - It is a custom or vendor specific identifier. ← Needs to be verified for security purposes.
- Website that helped with me on this part: https://github.com/shoulderhu/gitbook-tryhackme/blob/master/linux-server-forensics.md

—----------

```
fred@ip-10-67-132-165:/var/log/apache2$ cat access.log.1 | grep -a GXWR
192.168.56.206 - - [20/Apr/2021:13:30:15 +0000] "GXWR / HTTP/1.1" 501 498 "-" "M
ozilla/5.0 (iPhone; CPU iPhone OS 12_2 like Mac OS X) AppleWebKit/605.1.15 (KHTM
L"
```

- 13:30:15 was the NMAP scan performed.

—----------

Persistence Mechanisms II

Linux command **sudo -i** lets me open a login root shell.

- Link that help me find this command: https://linuxcapable.com/sudo-command-in-linux-with-examples/

I went back to **cd /var/log/apache2**

- Looked at apache.log & apache.log.1 file logs. ← <u>This is where you don't find the username and hostname!</u>

```
root@ip-10-67-132-165:~/.ssh# cat authorized_keys
ssh-rsa AAAAB3NzaC1yc2EAAAADAQABAAABgQDYCKt0bYP2YIwMWdJWqF3lr3Drs3sS9hiybsxz9W6d
G6d15mg0SVMSe5H+rPM6VmzOKJaVpDjT1Ll5eR6YcbefTF2bMXHveyvcrzDxyZeWdgBs5u8/4DZxEN6f
q6IZRRftmrMgMzSnpmdCm8kvacgq3lIjLx/sKAlX9GqPIz09t0Rk5MB7zk3lg1wdTZxZwwCHPbZW7mGl
VcxNBB9wdbAmcvezscoF0i7v0tY8iCoFlrBysOMBMrEJji2UONtI/wrt7AvoK+gshiG7VTjZ2oQBacny
HRToXHxOZiSIbCQrJ6rCxa32QOGQNmAVIucqYjRbJedz0NbGq7M9B+hBmG/mdtsoGOXQKyzoUlAbulRX
jSVtManiUyq9im1HBHfuduiBrbfcOKz24NMT7RaIsPsZCUCpfHaT7S5XplQypAjkxABds8jod/TXcTYi
bdWE9scrUUidgCsPELQlKEfhhZ8+cyjbMCGNB5LOgieJSVk6D1JC97TaFNi4X9/9i2UA+L0= kali@ka
li
```

- After I entered root with **sudo -i**, I searched up what authorized keys are and where they are located in Linux.
    - Website that helped me find authorized_keys file: https://linuxvox.com/blog/authorized_keys-linux/

```
root@ip-10-67-132-165:~# cd ~/.ssh
root@ip-10-67-132-165:~/.ssh# ls
authorized_keys
```

- I got this from the website and I then did **cat authorized_keys** to open the file. 'kali@kali' is the answer and was the username and hostname combination found.

—---------

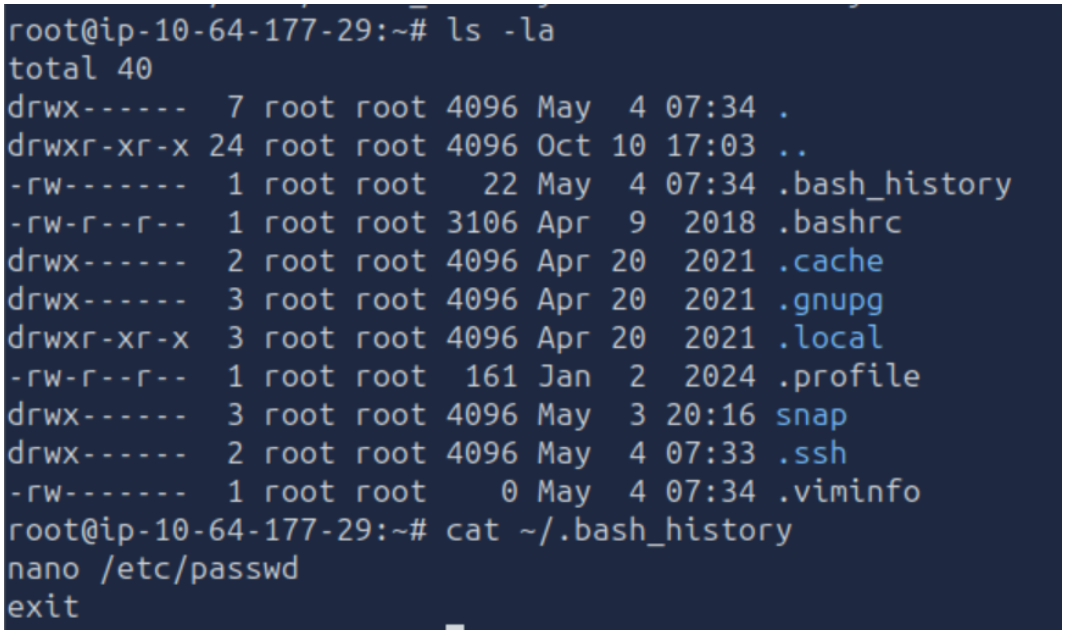
Program Execution History

Linux Program Execution History Sources:

1. **bash_history** - Contains a history of commands run in bash; this file is well known, easy to edit and sometimes disabled by default.
2. **auth.log** - Contains a history of all commands run using sudo.
3. **history.log (apt)** - Contains a history of all tasks performed using apt - is useful for tracking programme installation and removal.

systemd services also keep logs in the journald system.

- These logs are kept in a binary format and have to be read by a utility like **journalctl**.



- This took me a minute because I couldn't find the '.bash_history' file.
- I forgot Linux command **ls -la** can see hidden files in the home directory.
- This website reminded me to check hidden files: https://unanswered.io/guide/where-is-the-bash-history-file-located

—----------

Persistence Mechanisms III:

systemd services are defined in .service files which can contain:

1. The command that runs whenever the service starts
2. The user the service runs as
3. An optional description

Running **systemctl** will list all of the services loaded into the system. And much like Windows, there's usually a lot of them. It might be worth adding **--type=service --state=active** to the command as it will reduce the list to services that are running. Once the name of a suspicious service is found, more information can then be extracted by running **systemctl status <service name>** .

- Type **systemctl --type=service --state=active** to see less running services listed
  - https://linuxvox.com/blog/listing-linux-services-with-systemctl/ ← Systemctl help
- I am investigating malware on a Linux account.

```
 lvm2-monitor.service                loaded active exited  Monitoring of LVM2 mirrors, snapshots etc. using
 lvm2-pvscan@259:4.service           loaded active exited  LVM event activation on device 259:4
 lxcfs.service                       loaded active running FUSE filesystem for LXC
 ModemManager.service                loaded active running Modem Manager
 multipathd.service                  loaded active running Device-Mapper Multipath Device Controller
 networkd-dispatcher.service         loaded active running Dispatcher daemon for systemd-networkd
 polkit.service                      loaded active running Authorization Manager
 rc-local.service                    loaded active exited  /etc/rc.local Compatibility
 rsyslog.service                     loaded active running System Logging Service
 serial-getty@ttyS0.service          loaded active running Serial Getty on ttyS0
 setvtrgb.service                    loaded active exited  Set console scheme
 snapd.apparmor.service              loaded active exited  Load AppArmor profiles managed internally by sna
 snapd.seeded.service                loaded active exited  Wait until snapd is fully seeded
 snapd.service                       loaded active running Snap Daemon
 ssh.service                         loaded active running OpenBSD Secure Shell server
 systemd-journal-flush.service       loaded active exited  Flush Journal to Persistent Storage
 systemd-journald.service            loaded active running Journal Service
 systemd-logind.service              loaded active running Login Service
 systemd-modules-load.service        loaded active exited  Load Kernel Modules
 systemd-networkd-wait-online.service loaded active exited  Wait for Network to be Configured
lines 20-42
```

- Some of the active services in the Linux account during live malware.

```
WARNING!: THE RAM FLANGES ARE IN THE OFF POSITION SAFE OPERATION OF RAM DRIVER
IS NOT GUARANTEED!


ATTENTION!: THE BITBUCKET IS ALMOST FULL


WARNING!: DIHYDROGEN MONOXIDE DETECTED IN ATMOSPHERE


INFO!:  RETICULATING SPLINES


調試！: 如果發生堆棧衝突，則無法啟用堆棧！！！
```

- Some of the malware that has different messages as new ones arrive in the command prompt.

Tool to help for investigating malware on Linux:

- https://linuxvox.com/blog/scan-linux-for-malware/

—----------

```
-rw-r--r-- 1 fred fred 35802 Oct 10 17:21 'ystemctl --type=service --state=active'
```

- I saw this when I did ls -la?

Systemctl status Dbus.service
networkd-dispatcher.service
Systemd-networkd-wait-online.service

- These above did not help.

—----------

```
systemctl status IpManager
● IpManager.service
     Loaded: loaded (/etc/systemd/system/IpManager.service; enabled; vendor preset: enabled)
     Active: active (running) since Sat 2026-10-10 17:16:52 UTC; 37min ago
   Main PID: 1471 (bash)
      Tasks: 2 (limit: 4461)
     Memory: 3.3M
     CGroup: /system.slice/IpManager.service
             ├─1471 /bin/bash /etc/network/ZGtsam5hZG1ua2Fu.sh
             └─2602 sleep 10

Oct 10 17:53:38 ip-10-64-147-36 bash[2582]: tput: No value for $TERM and no -T specified
Oct 10 17:53:48 ip-10-64-147-36 bash[2584]: tput: No value for $TERM and no -T specified
Oct 10 17:53:48 ip-10-64-147-36 bash[2586]: tput: No value for $TERM and no -T specified
Oct 10 17:53:58 ip-10-64-147-36 bash[2589]: tput: No value for $TERM and no -T specified
Oct 10 17:53:58 ip-10-64-147-36 bash[2591]: tput: No value for $TERM and no -T specified
Oct 10 17:54:08 ip-10-64-147-36 bash[2595]: tput: No value for $TERM and no -T specified
Oct 10 17:54:08 ip-10-64-147-36 bash[2597]: tput: No value for $TERM and no -T specified
Oct 10 17:54:18 ip-10-64-147-36 bash[2599]: tput: No value for $TERM and no -T specified
Oct 10 17:54:18 ip-10-64-147-36 bash[2601]: tput: No value for $TERM and no -T specified
Oct 10 17:54:24 ip-10-64-147-36 systemd[1]: /var/lib/network//IpManager.service:1: Unknown section 'UNIT'. Ignor
```

- GitHub write-up saw IpManager.
- File path **/etc/network/ZGtsam5hZG1ua2Fu[.]sh** found.

```
fred@ip-10-64-147-36:~$ cat
WARNING!: THE RAM FLANGES ARE IN THE OFF POSITION SAFE OPERATION OF RAM DRIVER
IS NOT GUARANTEED!

/etc/network/ZGtsam5hZG1ua2Fu.sh
##[gh0st_1n_the_machine]
##
declare -a error_messages
error_messages[1]='ATTENTION!: THE BITBUCKET IS ALMOST FULL'
error_messages[2]='ACHTUNG!: DAS KOMPUTERMASCHINE IS NICHT GUD'
error_messages[3]='WARNING!: THE RAM FLANGES ARE IN THE OFF POSITION SAFE OPERATION OF RAM DRIVER IS NOT GUARANTEE
D!'
error_messages[4]='ERROR!: THE STACK ARRANGER IS NOT ENABLED BEWARE OF STACK COLLISIONS'
error_messages[5]='調試!: 如果發生堆棧衝突，則無法啟用堆棧!!!'
error_messages[6]='INFO!:  PURGING RAM BITS'
error_messages[7]='INFO!:  NODE GRAPH OUT OF DATE REBUILDING'
error_messages[8]='INFO!:  RETICULATING SPLINES'
error_messages[9]='WARNING!: DIHYDROGEN MONOXIDE DETECTED IN ATMOSPHERE'
error_messages[10]='INFO!: VENTING OXYGEN'
error_messages[11]='WARNING!: /dev/null IS 95% UTILIZED'
error_messages[12]='METTERE IN GUARDIA!: LE FLANGE DEL RAM SONO IN POSIZIONE OFF IL FUNZIONAMENTO SICURO DEL RAM D
RIVER NON È GARANTITO!'
```

- This shows the messages of the malware
- Flag is [gh0st_1n_the_machine]